MIL-P-8945A
23 August 1977
SUPERSEDING
MIL-P-8945(ASG)
9 August 1967
MIL-P-4977(USAF)
27 July 1955
MIL-C-5427A(ASG)
10 April 1958

MILITARY SPECIFICATION

PANELS, RELAY, PILOT, ENCLOSED, AIRPORT LIGHTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for airport power distribution relay panels.

1.2 Classification. Relay panels covered by this specification shall be of the following types, as specified (see 6.2):

Type I - 24 relays, all double-pole, single-throw (MS21961)

Type II - 24 relays, 16 double-pole, single-throw, and 8 relays, double-pole, double-throw (MS21961)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices

FSC 6110

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-686	Cable and Cord, Electrical; Identification Marking and Color Coding of
MIL-STD-808	Finish, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS21961	Panels, Relay, Pilot, Enclosed, Airport Lighting (Types I and II)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION

NEMA STD. NO. IC-1-1965 Industrial Controls

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 First article. The relay panels furnished under this specification shall be a product which has been subjected to and which has passed the first article inspection and tests specified herein.

3.2 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in 3.2.1.

3.2.1 Use of commercial parts. Bolts, nuts, washers, cotter pins, lock rings, and similar fastening devices used for assembly may be selected from commercial sources, provided military or other standard parts are not specifically called out by this specification or associated standards. If such commercial parts are used, they shall be replaceable by military standard (MS) parts without alterations. The corresponding MS part numbers are to be referenced in the parts list and on the contractor's drawings.

3.3 Materials

3.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Gaskets. All gaskets and other materials used in the fabrication shall be of a type which will not emit smoke or vapor during rated operation of the relay panel.

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3.3.3 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. When used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a nermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4 Design and construction. The design and construction of the panel shall be in accordance with MS21961. All threaded parts shall be suitably locked.

3.4.1 Design conditions. Each panel shall be designed for continuous operation at an ambient temperature ranging from -20° to +55°C and voltage ranging from 90 to 125 volts, 60 Hertz (Hz).

3.5 Components. The relay panels shall consist of the following major components:

a.	Enclosure	(3.5.1)
b.	Mounting plate	(3.5.2)
c.	Grounding lug	(3.5.3)
d.	Relays	(3.5.4)
e.	Terminal boards	(3.5.5)
f.	Wiring	(3.5.7)

All components shall be commercial parts modified to meet the requirements specified herein.

3.5.1 Enclosure. The panels shall have a type I enclosure in accordance with NEMA Std. No. IC-1-1965. The enclosure shall be completely rigid and self-supporting and shall be designed for surface mounting.

3.5.1.1 The enclosure, including the door and trim, shall be fabricated of sheet steel in accordance with MS21961.

3.5.1.2 Dimensions. The dimensions of the enclosure shall be as specified on MS21961. Dimensions and tolerances not specified shall be consistent with the best shop practices. Where dimensions or tolerances may affect the interchangeability of component parts, they shall be held or limited accordingly.

3.5.1.3 Knockouts. The enclosure shall have three concentric knockouts on each end and four concentric knockouts on each side. The knockouts shall be symmetrically positioned as specified on MS21961.

3.5.1.4 Door and trim. The door trim shall be of one piece and in accordance with the dimensions specified on MS21961. The trim shall be held to flange provided on the enclosure by sufficient trim clamps to ensure rigidity with a tight fit between the trim and enclosure. The door shall be flush with the trim. Hardware for the door shall consist of flush hinges and a flush spring latch with a ring handle. The hinges shall be mounted on the right side of the door. A neat molding shall completely surround the door in order to provide reinforcement and a rabbet around the door opening. The molding shall be mitered around the corners.

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3.5.1.5 Mounting holes. Four mounting holes shall be provided as shown on MS21961.

3.5.1.6 Finish. Cleaning and finishing shall be in accordance with MIL-STD-808, finish F-100KG.

3.5.2 Mounting plate. A removable, steel mounting plate shall be provided for mounting the relays and terminal strips. The dimensions of the mounting plate shall be as shown on MS21961. The plate shall be drilled and tapped to facilitate mounting the relays and terminal strips. The steel plate shall be mounted 0.25 inch from the back of the enclosure and shall be positioned as shown on MS21961. The relays and terminal strips shall be arranged to the steel plate to provide a wiring gutter at either side and at the top end of the enclosure. The general arrangement of the relays and terminal strips shall be as shown on MS21961.

3.5.3 Grounding lug. A grounding lug, suitable for attachment of the grounding wire (No. 8 AWG), shall be provided inside the enclosure.

3.5.4 Relays. Standard commercial relays shall be mounted inside the cabinet on a mounting plate as specified on MS21961. Single-throw relays shall be the normally open type. The double throw relays shall have two normally open and two normally closed contacts. The relays shall be low-burden, heavy-duty, industrial control relays. The relay coils shall be rated at 115V, 60 Hz. The inrush current of the coil shall not exceed 0.3 amp. and the continuous holding current of the coil shall not exceed 0.15 amp. at rated voltage and frequency. The voltage for pull-in of the relays shall be between 80 and 90V and for drop-out shall be between 55 and 65V. The relay contacts shall have a continuous rating of 250V, 60 Hz, 10 amp. for resistive loads. The rupture rating of the contacts shall be 250V, 45 amp. make, 45 amp. break, for resistive loads. Moving parts of the relay shall be removable from the front. The moving contacts shall be spring loaded in the closed position, provide positive contact and wiping action. The stationary contacts shall be rigidly mounted. The relay terminals for external connection shall be of the screw type (solder-type terminals shall not be acceptable). The center-to-center spacing between adjacent terminals and between adjacent stationary contact shall be at least 0.75 inch. The air gap of the contacts in the open position shall be a minimum of 0.075 inch. The relays shall be arranged in six horizontal rows with four relays in each row. The relays shall be mounted in a vertical position with the contacts at the top of the relay. The relay-to-relay and relay-to-terminal strip spacing shall be adequate to permit wiring in accordance with good commercial practice.

3.5.5 Terminal boards. Screw-type terminals boards, shall be double-row, linked, front-opening, barrier type, rated at 20 amp., 600V, to accommodate wires from No. 20 to No. 10 AWG, using No. 8-32, or larger, terminal screws. Terminal boards shall be fabricated of plastic conforming to MIL-M-14, type CFG. Buchanan boards No. 515, or equal, are considered satisfactory. Terminals shall be marked with the terminal designations as shown on MS21961. Terminal markings shall be placed over the terminals designated. Terminal boards shall be mounted on the mounting plate as shown on MS21961. The input terminal blocks shall be located at the left of the relays and the output terminal blocks shall be located on the right of the relays.

3.5.6 Marking. Relays shall be permanently marked and designated as A, B, C, D, et cetera, as shown on MS21961. Relay A shall be in the upper left corner and marking shall proceed from left to right on each succeeding row of relays. Control terminals shall also be marked A, B, C, D, etc., reading from left to

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right. The last control terminal shall be marked LN. Input terminals, reading from top to bottom, shall be marked A1, A2, B1, B2, C1, C2, et cetera. Output terminals, reading from top to bottom shall be permanently marked A3, A4, B3, B4, C3, C4, et cetera. Marking shall be accomplished on marking strips not less than 0.375-inch wide provided for that purpose. Sufficient space shall be reserved (over 50 percent of width) on the marking strip at each terminal of the terminal strips to permit identification of incoming wires. The letters I and O shall not be used in marking relays and terminals. The mounting plate adjacent to each terminal strip shall be marked:

Upper terminal block - Control terminals
Left terminal block - Input terminals
Right terminal block - Output terminals

The terms right and left refer to the operator facing the panel.

3.5.6.1 Ground. The mounting plate shall be marked ground adjacent to the grounding lug.

3.5.7 Panel circuitry. There shall be continuity between input and output terminals only as specified below:

Type I panels - With 115V, 60 Hz applied between control terminal LN and each of the other control terminals in turn (A to Z inclusive), the correspondingly marked relay shall close, and there shall be continuity between the input and the output terminals corresponding to the energized relay as follows:
between terminals 1 and 3, and
between 2 and 4.

Type II panels - (a) With no voltage applied to any of the control terminals, there shall be continuity between the correspondingly marked input and output terminals S1 and S5, S2 and S6 to Z1 and Z5, Z2, and Z6, inclusive. (b) With 115V, 60 Hz applied between control terminal LN and each of the other control terminals in turn (A to Z inclusive), the correspondingly marked relay shall close, and there shall be continuity between the input and output terminals corresponding to the energized relay as follows:
between terminals 1 and 3, and
between 2 and 4.

3.5.8 Wiring. The relay panel shall be completely wired in accordance with MS21961. Bundles of wires shall be compact and neatly tied. All wires shall be flame retardant, synthetic, stranded, 600V, No. 12 AWG (National Electric Code, type T). Connections shall be as shown on MS21961, and meet the continuity requirements of 3.5.7. Wiring shall be color coded in accordance with MIL-STD-686.

3.5.8.1 Wiring diagram. An 8- by 8-inch plastic or metal plate shall be mounted on the back of the enclosure door. The plate shall have a wiring diagram of the internal connections between the terminal blocks and relays permanently marked. The wiring diagram shall be as shown on MS21961. The wiring diagram shall be identified and oriented in a manner which will facilitate its use.

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3.6 Performance. When tested as specified, the relays and relay panels shall meet all requirements without mechanical or electrical damage or operational malfunctioning.

3.6.1 Relay insulation resistance. When tested as specified in 4.6.6.1, the insulation resistance shall be not less than 100 megohms.

3.6.1.1 Pull-in voltage. When tested as specified in 4.6.4, pull-in voltage shall be 85 ±5 volts. Contacts shall not assume an intermediate position.

3.6.1.2 Dropout voltage. When tested as specified in 4.6.5, dropout voltage shall be 60 ±5 volts. Contacts shall not assume an intermediate position.

3.6.1.3 Coil current. When tested as specified in 4.6.6.2, the current shall not exceed 0.15 amp.

3.6.1.4 Inrush coil current. When tested as specified in 4.6.6.2.1, the inrush coil current shall not exceed 0.30 amp.

3.6.1.5 Contact drop. When tested as specified in 4.6.6.3, the contact drop shall not exceed 0.1V.

3.6.1.6 Rupture. When tested as specified in 4.6.6.4, there shall be no failure or damage to the relays.

3.6.1.7 Continuous operation. When tested as specified in 4.6.6.5, there shall be no electrical or mechanical damage to the relays and the contact drop shall not exceed 0.1V at the conclusion of the tests.

3.6.1.8 Relay operating cycles. When tested as specified in 4.6.6.6, there shall be no mechanical or electrical damage to the relay during or at the conclusion of the tests.

3.6.2 Panel circuitry. When tested as specified in 4.6.2, all components shall operate properly and all contacts shall make and break properly.

3.6.3 Panel dielectric. When tested as specified in 4.6.3, there shall be no electrical or mechanical damage to the panel or any of its components.

3.7 Interchangeability. All parts functionally and mechanically interchangeable shall be identified by one part number. The drawing number requirements of MIL-STD-100 shall govern the changes in the manufacturer's part number.

3.8 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9 Workmanship. The relay panels, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alignment of parts and tightness of assembly screws and bolts.

3.9.1 Cleaning. The relay panels shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign materials shall be removed during and after final assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Test conditions. Unless otherwise specified (see 6.2), the tests specified herein shall be performed at room temperature (approximately 25°C).

4.4 First article inspection

4.4.1 Test sample. One panel of each type shall be subjected to the inspection and tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the first article inspection and tests, a test report shall be prepared in accordance with MIL-STD-831.

4.4.3 First article tests. The first article tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance inspection. Each relay panel shall be subjected to a quality conformance inspection consisting of the individual tests specified under 4.5.1.

4.5.1 Individual tests. Each relay panel shall be subjected to the following tests as specified under 4.6.

a. Examination of product	See 4.6.1
b. Panel circuitry	See 4.6.2
c. Panel dielectric	See 4.6.3
d. Pull-in voltage	See 4.6.4
e. Dropout voltage	See 4.6.5.

4.6 Test methods

4.6.1 Examination of product. Each relay panel shall be inspected to determine compliance with the requirements specified herein with respect to design, materials, workmanship, and marking.

4.6.2 Panel circuitry. Each relay panel shall be subjected to circuitry tests to determine compliance with the requirements of 3.5.7 and 3.6.2.

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4.6.3 Panel dielectric. Each relay panel shall be subjected to a dielectric test conducted at 1,500V ac or 2,000V dc to determine that each terminal, and the wires connected thereto, is properly insulated from ground and from all other wires.

4.6.4 Pull-in voltage. The pull-in voltage shall be determined by connecting the coil of each relay to a continuously variable, 60 Hz potential source and gradually increasing the voltage to close the normally open contacts. Pull-in voltage shall be 85 ±5V. An indicating device shall be used to determine that the contacts operate satisfactorily. The contacts shall not assume an intermediate position for any applied voltage.

4.6.5 Dropout voltage. The dropout voltage shall be determined by connecting the coil of each relay to a continuously variable, 60 Hz potential source and gradually decreasing the voltage from 115V until the normally open contacts are open. Dropout voltage shall be 60 ±5V. An indicating device shall be used to determine when the contract open. The contract shall not assume an intermediate position between fully closed and fully open as the voltage is reduced.

4.6.6 Relays. Four relays shall be selected at random from the relay panel submitted for qualification and subjected to:

4.6.6.1 Insulation resistance. The relay shall be tested as follows:

a. A potential of 2,000V dc shall be applied between the coil terminals and grounded metal parts for 1 minute; the insulation shall meet the requirements of 3.6.1.

b. With the relay contacts in the deenergized position, a potential equal to twice the rated contact voltage plus 2,000V dc shall be applied between nonconnected contacts and between all other contacts and grounded metal parts for 1 minute. This procedure shall be repeated with the contacts in the energized position. The insulation resistance shall meet the requirements of 3.6.1.

4.6.6.2 Coil current for holding. With the relay stabilized at room temperature (25° ±5°C), 115V at 60 Hz shall be applied to the coil terminals, and the current flow determined 10 seconds or longer after application of coil voltage.

4.6.6.2.1 Inrush coil current. With the relay stabilized as above, the inrush coil current at 115V, 60 Hz, shall be determined. The root means square current shall not exceed 0.30 amp.

4.6.6.3 Contact drop. The voltage drop across relay contacts, as measured at the relay terminals, shall be determined at 10 amperes resistive load through the contacts when the coil is energized at 65V at 60 Hz. The drop shall not exceed 0.1V as determined from an average of 10 consecutive measurements.

4.6.6.4 Rupture. The relay contacts shall be made to make and break 45 amp. at 250V, 60 Hz for 50 operations. The on period shall be 0.2 ±0.005 second, and the off period shall be 30.0 seconds. Following this test, the relay shall be subjected to and pass the test specified in 4.6.6.3.

4.6.6.5 Continuous operation. The relays shall be subjected to room temperature (25° ±5°C) for 100 hours, at 115V, 60 Hz, across the coil and with

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a resistive load of 10 amp. at 250V throughout the contacts. Immediately after this test, the relay shall be subjected to and pass the test specified in 4.6.6.3.

4.6.6.6 Relay operating cycles. The relay shall be subjected to 20,000 operating cycles at resistive load and voltage. After this test the relay shall be subjected to and pass the test specified in 4.6.6.3.

4.7 Packaging inspection. The relay panel shall be examined to determine that the preservation, packing, packaging, and marking is in conformance with section 5.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. All items of the equipment shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or order.

6. NOTES

6.1 Intended use. The relay panels covered by this specification are to be used for controlling and interlocking airport lighting circuits when the airport lighting control panels are located at an excessive distance from the vault containing the primary coil switches and contactors. It is also intended for direct control of a few small loads, such as the rotating beacon, and wind direction indicator lights.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. MS part number of panel (see 1.2)

c. Test conditions, if other than as specified (see 4.3)

d. The level of preservation, packaging, and packing required (see 5.1).

6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirement List (DD Form 1423) and incorporated into the contract. Data produced by this specification is the test report specified in 4.4.2.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulation (ASPR). The first article should be a first production item. The procuring activity should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Hazards guarantee. It should be requested that the bidder submit proof to the procuring activity that the relay assembly specified in 3.5 which he proposes to supply under this specification conforms to the standards of the Underwriters' Laboratories, Inc., as regards fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc. will be accepted as evidence of conforming to the specification requirements.

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6.5.1 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the procuring activity that this panel conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc. Compliance with the fire and casualty hazards requirements does not absolve the bidder from complete compliance with the other requirements of this specification.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 11

Reviewer activities:
Navy - AS
Air Force - 99
DSA - GS

Preparing activity:
Air Force - 11

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